RTA-SUM - Timing Client Service (TCS)

Design Document

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1 Introduction and Goals

This document intends to describe the design of the SUM_TCS module. The scope of this document is to capture the implementation based on the requirements as defined in SDV-8002 AUTOSAR Standard Utility Module Specification. For a better process and higher quality design, this can be seen as an extension to the before mentioned document.

1.1 Introduction

1.1.1 System Purpose

This document defines a set of standard software compositions which are referred to as GM Standardized Utility Modules (SUM). Use of the SUM modules allows for standardized specification of behavior requirements and expectations with respect to the interface to the software infrastructure (BSW Services). This allows functional software to be completely abstracted from the software infrastructure. SUM will be used to communicate a set of standardized implementations of the interactions between SW-C and BSW.

The SUM_TCS is a software module in the application layer of the standardized architecture of AUTOSAR. It is primarily intended to be used to provide a synchronized time stamp for the extended date record of a DTC.

SUM_TCS contains the following functions:

- Receive the global time from the Master Clock global time provider via serial data communication
- Synchronize the local ECU time to the master global time
- Provide the last received value of the Master Clock signal (default if never received) following initialization, until the initial Master Clock signal is received via serial communication
- Keep a local timer source to maintain ECU internal time source between reception events of the Master Clock signal.
- Offer an interface to the Dcm and Dem modules, providing synchronized timestamp.

1.1.2 Document Purpose

This document introduces the key ideas of the architecture of SUM_TCS. Its intended audience comprises System Architects, Product Architects, Developers and Testers who need a rough understanding on the concepts of the product. The document describes the product and its functionality on the component level, the interaction and communication between those components and the integration of ETAS wide shared components. The document does not explain the components design in detail.

1.2 Goals of the Architecture

1.2.1 Goals

The main goals of the architecture are:

1. **Configurability**: The module has to be designed with the highest possible level of configurability, in order to be easy to adapt it in the future to changes on the standard or even in some project requirements.
2. **Maintainability**: The architecture of the component shall be clearly defined and easy to understand, in order to guarantee the maximum possible level of maintainability.
3. **Performance**: The component shall be designed in order to maximize run-time execution speed as well as minimize memory consumption.
4. **Correctness of Implementation**: The Detailed Design must to be aligned to the Architecture document (SDV-8002).

1.3 Stakeholders

Stakeholders for SUM_TCS module are GM, Bosch, Systems engineers, Software architect, QA personnel.

1.4 Definitions, Acronyms and Abbreviations

Term/Abbreviation	Definition
API	Application Program Interface
BSW	Basic Software
BswM	Basic Software Mode Manager
CAN	Controller Area Network
CR	Change Request
Dcm	Diagnostic Communication Module
DTC	Diagnostic Trouble Code
DIAG	Diagnostic Application Services
ECU	Electronic Control Unit
EcuM	ECU State Manager
ERRH	Error Handling Module
GM	General Motors
LIN	Local Interconnect Network
LNM	LIN Network Manager
NVRAM	Non Volatile Random Access Memory
NvM	NVRAM Manager
OAW	Open Architecture Ware
PMS	Power Mode Slave
PNC	Partial Network Coordinator
RTE	Run Time Environment
SOH	State Of Health Manager

Term/Abbreviation	Definition
SSM	Signal Status Monitor
SSC	Security Service Coordinator
SUM	Standard Utility Module
SUSD	Startup Shut down Coordinator
SW	Software
SW-C	Software Component
TCS	Timing Client Service
WdgM	Watchdog Manager

1.5 Referenced Documents

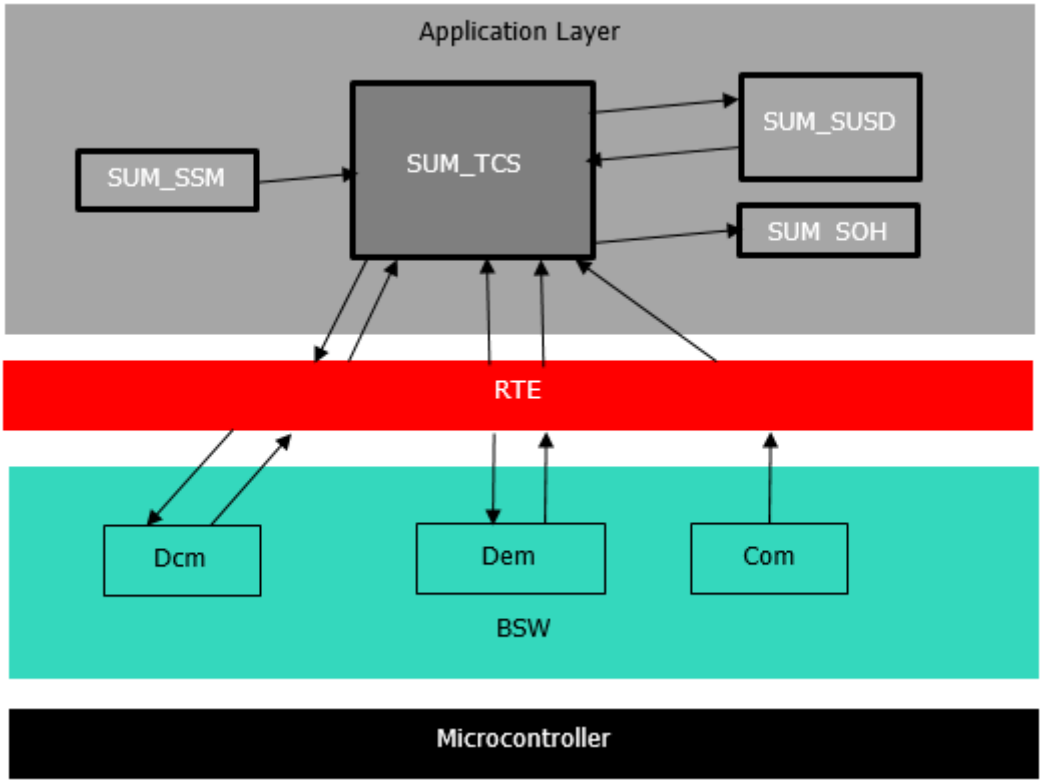
Document Name	Description	Version	Released
SDV-8002 AUTOSAR Standard Utility Modules Specification	TCS module specification	27.27.174.3p1	02/15/2024
SDV-6001 Diagnostic Infrastructure Specification	TS Slave Requirements	27.27.174.3p1	02/15/2024

2 Architecture Constraints

2.1 Environment

Tool Name	Version	Description
ISOLAR-AB	12.3.3BETA1	AUTOSAR Authoring and BSW Configuration tools.
RTA-BSW	12.3.3pr1	BSW automatic configuration and code generation.
RTA-RTE	12.4.0	RTE generator used for testing.
MemMapGen	12.3.0	Memory Map file generator.
CANTATA	9.5	Used for functional testing.

2.2 Layers Definition



The applied system architecture is based on Autosar v4.0.3. For detailed information on it refer to Autosar.org website.

2.3 Generic component interface

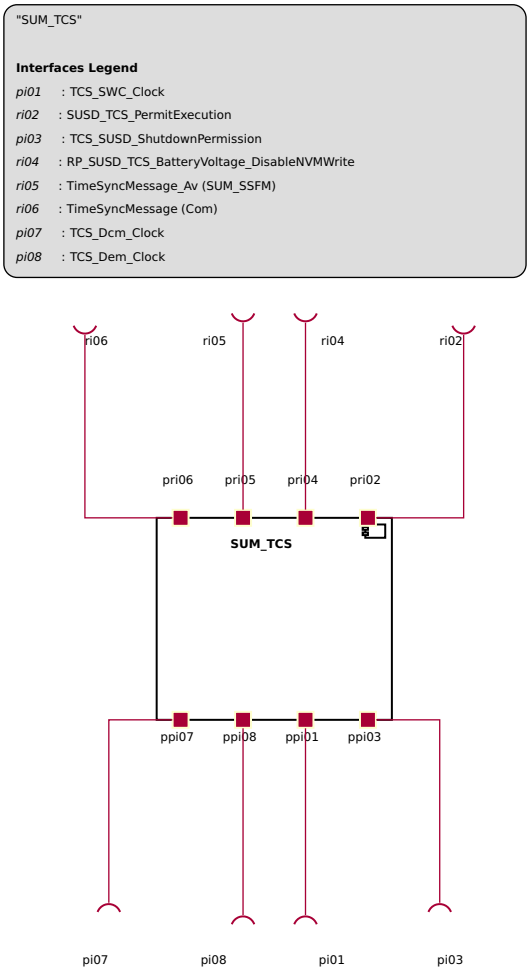


Diagram 2.1: Generic Component Interface

2.3.1 Port Interfaces

[$SDD_TCS 023]

Port Interface	Port Type	P-Port	R-Port	Description
TCS_SWC_Clock	Sender / Receiver	SUM_TCS	SUM_SOH	Provides timestamp data for error monitoring.
TimeSyncMessage	Sender / Receiver	remote SW-C	SUM_TCS	Master Clock Signal received from TS_Master ECU.
TimeSyncMessage_Av	Sender / Receiver	SUM_SSM	SUM_TCS	Availability of the Master Clock Signal.
SUSD_TCS_PermitExecution	Sender / Receiver	SUM_SUSD	SUM_TCS	Informs SUM_TCS whether it is allowed to perform its functions.
TCS_SUSD_ShutdownPermission	Sender / Receiver	SUM_TCS	SUM_SUSD	Informs SUM_SUSD whether SUM_TCS is ready for a shutdown.
TCS_Dem_Clock	Client / Server	SUM_TCS	Dem	Provides Dem with timestamp data to write in the extended data record of a DTC.
TCS_Dcm_Clock	Client / Server	SUM_TCS	Dcm	Used to implement the Read function of DID F129.
RP_NvMService_TCSTimeValue	Client / Server	NVM	SUM_TCS	Needed to use NVM services to store `SumTCSInitValue`.
RP_SUSD_TCS_BatteryVoltage_ DisableNVMWrite	Sender / Receiver	SUM_SUSD	SUM_TCS	Needed to read battery voltage condition to perform NvmWrite.

[$Satisfies $SRS_TCS 009] [$Satisfies $SRS_TCS 012] [$Satisfies $SRS_TCS 018] [$Satisfies $SRS_TCS 031]

2.3.2 TCS DID services to Dcm

[$SDD_TCS 025]

DID Name	ID	Range	Access	Default Value	Description
Total Vehicle Time Value	0xF129	0 - 1,099,511,627,775 ms	Read	0 ms	Total Vehicle Time Value is a running time counter that is received by the Time Leader. This counter is locally incremented by Time Followers, when the signal is not available, and is also used for Time Stamping.

[$Satisfies $SRS_TCS 024] [$Satisfies $SRS_TCS 025]

2.4 Other constraints and assumptions

2.4.1 Assumptions

1. The common interfaces among SUMs and SWCs are defined in `GM_SUM_Interfaces.arxml`.
2. The ports and interfaces unique to SUM_TCS shall be provided in `SUM_TCS_Cfg_SWCD.arxml`.
3. The resolution of a `TimeSyncMessage` is 100 ms. This is equal to saying that the timestamp in the `TimeSyncMessage` is a counter representing the number of 100 ms increments. It follows that the 36 bits of the `TimeSyncMessage` can represent up to $2^{36} \cdot 100$ ms (approximately 6.872E9 seconds, 79536 days, 217 years).

2.4.2 Constraints

Internal Data Types: Special design considerations have been taken into account due to the configurability required to implement SUM_TCS. Data types which are not defined exclusively in the RTE are used in order to facilitate the configuration required for this module.

Non-Volatile Memory Access: In order to fulfill the requirement to store the `SumTCSInitValue` in memory, a RAM mirror block is used for storing this data in non-volatile memory. It is assumed the user will configure the NVM stack to support this required information. Further, it is expected that during shutdown, the appropriate BSW services will write that specific NVM block to persistent storage. The operation should be performed only if the battery voltage is good after checking it from SUM_SUSD.

Exlusive Areas: In order to prohibit concurrent data access, SUM_TCS configures each runnable to use the `RunsInsideExclusiveArea` protection.

3 Building Block View (Static View)

The SUM_TCS module is a SWC within the context of AUTOSAR. It consists of static files and generated files.

3.1 Configuration

SUM_TCS configuration is done the same way as for standard BSW modules. A parameter definition file is included with the module which allows the users to configure the available parameters. The generated files change based on the configuration. The configuration parameters for TCS are provided in the paramdef file `SUM_TCS_EcuParamDef.arxml`. The user shall configure values to the various parameters provided in the paramdef: based on the inputs configured, TCS shall use OAW scripts to generate the source and header files along with the SWCD.

3.1.1 Included Containers:

Name	Multiplicity	Scope / Dependency
TcsGeneral	1..1	General configuration paramers for TCS

3.1.2 TcsGeneral

[$SDD_TCS 031]

Parameter Name	`MasterClockPollingPeriodicity`
Description	Time in msec for the periodicity of MasterClockPolling
Multiplicity (upper..lower)	1..1
Category	`ECUC-INTEGER-PARAM-DEF`
Range	1..1000
Default value	100

Parameter Name	`PermitExecutionMask`
Description	Bitmask used to identify the Wakeup bit in the `SUSD_TCS_PermitExecution` interface.
Multiplicity (upper..lower)	1..1
Category	`ECUC-INTEGER-PARAM-DEF`
Range	0..255
Default value	254

Parameter Name	SumTCSGlobalTimePrecision
Description	Local time source accuracy in msec. Must be a positive multiple of 100 ms and be greater than or equal to TimeManagementPeriodicity.
Multiplicity (upper..lower)	1..1
Category	ECUC-INTEGER-PARAM-DEF
Range	1..1000
Default value	100

Parameter Name	TimeManagementPeriodicity
Description	Time in msec for the periodicity of TimeManagement
Multiplicity (upper..lower)	1..1
Category	ECUC-INTEGER-PARAM-DEF
Range	1..1000
Default value	100

Parameter Name	TcsGenerateInitRunnable
Description	Enable or Disable the Runnable Init feature.
Multiplicity (upper..lower)	1..1
Category	ECUC-BOOLEAN-PARAM-DEF
Range	NA
Default value	false

Parameter Name	TcsInitFuncPeriodicity
Description	Specifies time in ms for the periodicity of the runnable Runnable_TCS_Init.
Multiplicity (upper..lower)	0..1
Category	ECUC-FLOAT-PARAM-DEF
Range	1.0..5000.0
Default value	1000.0

Parameter Name	TCS_TimeSyncMessageSignalRef
Description	Reference to the ComSignal or ComGroupSignal used for the TimeSyncMessage reception. Determines the signal type used for master clock polling.
Multiplicity (upper..lower)	1..1
Category	ECUC-CHOICE-REFERENCE-DEF
Range	-..-
Default value	-

Parameter Name	TCS_ServiceBased_TimeSyncMessage_Ref
Description	Reference to the implementation data type element for service-based TimeSyncMessage communication. When configured, TCS uses SOME/IP service-based reception instead of signal-based.
Multiplicity (upper..lower)	0..1
Category	ECUC-FOREIGN-REFERENCE-DEF
Range	-..-
Default value	-

Parameter Name	TCS_ServiceBased_TimeSyncMessage_DataTypeMappingSet
Description	Reference to the data type mapping set required for service-based TimeSyncMessage communication. Must be configured when TCS_ServiceBased_TimeSyncMessage_Ref is set.
Multiplicity (upper..lower)	0..1
Category	ECUC-FOREIGN-REFERENCE-DEF
Range	-..-
Default value	-

Parameter Name	DID_F129_TotalVehicleTimeValue_Reference
Description	Reference to DCM DID 0xF129 Total Vehicle Time Value
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	-..-
Default value	-

Parameter Name	TcsDemTimestampRef
Description	Reference to Dem Timestamp
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	-..-
Default value	-

[$Satisfies $SRS_TCS 019]

3.2 Calibration Parameters

N/A

4 Runtime View (Dynamic View)

4.1 System Initialization

[$SDD_TCS 017]
The API shall perform TCS initialization.
[$Satisfies $SRS_TCS 020]

[$SDD_TCS 018]
TCS will have an initialization routine invoked by BswM or initialization runnable and this is decided by the configuration parameter `TcsGenerateInitRunnable`. When the configuration parameter is set to TRUE, the initialization is done by the `Runnable_TCS_Init` that calls the static `TCS_Init`, otherwise the initialization is done by the global `TCS_Init`, called by the BswM Init rule.
[$Satisfies $SRS_TCS 021] [$Satisfies $SRS_TCS 020]

[$SDD_TCS 027]

Name	`TCS_Init`
In parameters	`void`
Out parameters	N/A
InOut parameters	N/A
Return	`void`
Description	This function is responsible for: • Initializing the output port to initial value • Transitioning the execution state of the SUM_TCS module

[$Satisfies $SRS_TCS 029]

4.2 System De-initialization

N/A

4.3 Detailed Design

4.3.1 Macros

[$SDD_TCS 028]

Name	`TCS_RESETREQUEST_MASK`
Value	`0x40`
Description	Bitmask used to identify the ECU Reset Requested flag bit in the `SUSD_TCS_PermitExecution` interface. The ECU Reset Requested flag is bit 6.

[$Satisfies $SRS_TCS 013]

[$SDD_TCS 029]

Name	TCS_ENABLE_SHUTDOWN
Value	255u
Description	Value corresponding to Enable for the TCS_SUSD_ShutdownPermission interface.

[$Satisfies $SRS_TCS 013]

[$SDD_TCS 030]

Name	TCS_SHUTDOWN_CNT_LIMIT
Value	4u
Description	Upper limit for the alive counter transmitted with the TCS_SUSD_ShutdownPermission interface command SUSD_UserRequest. This behaviour is described in the SUM_SUSD requirements in SDV-8002.

[$Satisfies $SRS_TCS 012]

4.3.2 Data Types

Name	TimeSyncMessageType
Type	uint64
Description	A 64-bit unsigned integer containing a TimeSyncMessage

4.4 Critical timing constraints

- The periodicity of Runnable_TCS_MasterClockPolling must be less than or equal to the transmission rate of the TimeSyncMessage received by the Time Leader.
- The periodicity of Runnable_TCS_TimeManagement must be a submultiple of the resolution of the TimeSyncMessage (100 ms).

4.5 Task management

4.5.1 Periodic Runnables

[$SDD_TCS 001]
Runnable_TCS_MasterClockPolling shall use a variable EA_LocalTimeClock to keep a local time source and synchronize with Master Clock signal.
[$Satisfies $SRS_TCS 001]

[$SDD_TCS 003]
Runnable_TCS_MasterClockPolling shall receive the MasterClockSignal.
[$Satisfies $SRS_TCS 003] [$Satisfies $SRS_TCS 027]

[$SDD_TCS 006]
SumTCSInitValue should be first initialized to zero.
[$Satisfies $SRS_TCS 006]

[$SDD_TCS 013]
Runnable_TCS_MasterClockPolling shall check the validity of the received MasterClockSignal based on the value of MasterClockSignalAvailability.
[$Satisfies $SRS_TCS 015]

[$SDD_TCS 019]
Runnable_TCS_MasterClockPolling shall implement the local time counter with same resolution as of the timestamp and synchronize the same with the reception of every valid Time Sync signal.
[$Satisfies $SRS_TCS 022] [$Satisfies $SRS_TCS 028]

[$SDD_TCS 020]
Runnable_TCS_MasterClockPolling should record the last clock local value into the variable SumTCSInitValue.
[$Satisfies $SRS_TCS 023]

[$SDD_TCS 026]
Runnable_TCS_MasterClockPolling shall consume the time sync signal.
[$Satisfies $SRS_TCS 030]

[$SDD_TCS 034]
Runnable_TCS_MasterClockPolling shall support multiple TimeSyncMessage reception modes: service-based (ComSignal) and signal-based (ComSignal or ComGroupSignal), determined by configuration parameters.

Name	Runnable_TCS_MasterClockPolling
Type	Fixed
Inter-runnable Variables	EA_LocalTimeClock EA_SubCounter
Managed Input Interfaces	MasterClockSignal MasterClockSignalAvailability
Managed Output Interfaces	N/A
Per Instance Memory	N/A
Return	void
Description	This runnable is responsible for: • Polling new Master Clock signal and availability signal • Synchronizing local timer to Master Clock The TimeSyncMessage reception mode is determined by the configuration: • **Service-based** (SOME/IP): when TCS_ServiceBased_TimeSyncMessage_Ref is configured and comSignalType is UINT8_N, the signal is read via a ComSignal as a service data element with transformer error handling. The received byte array is converted to uint64 via TCS_Prv_byteArray_to_uint64. • **Signal-based ComGroupSignal**: when TCS_TimeSyncMessageSignalRef references a ComGroupSignal with comSignalType UINT64, the signal is read via a signal group struct. When E2E is configured, the Rte_Read call includes a Rte_TransformerError parameter. • **Signal-based ComSignal** (default): when TCS_TimeSyncMessageSignalRef references a ComSignal with comSignalType UINT64, the signal is read directly as a uint64 value.

[$Satisfies $SRS_TCS 001] [$Satisfies $SRS_TCS 003]

[$SDD_TCS 011]
Runnable_TCS_TimeManagement shall increment the variable EA_SubCounter during each period of execution as long as the PermitExecution notification is active.
[$Satisfies $SRS_TCS 011]

[$SDD_TCS 015]
Runnable_TCS_TimeManagement shall update EA_LocalTimeClock to ensure it meets SumTCSGlobalTimePrecision.
[$Satisfies $SRS_TCS 016]

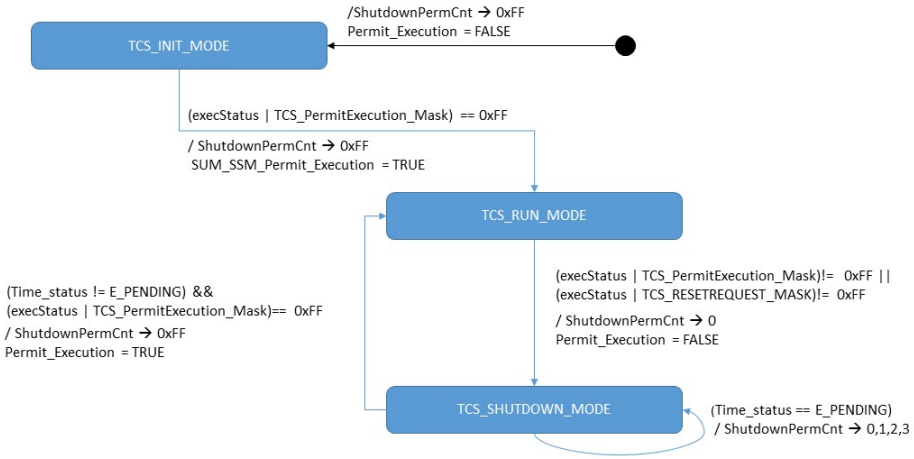
[$SDD_TCS 016]
EA_LocalTimeClock payload shall match the payload of the MasterClock signal.
[$Satisfies $SRS_TCS 022]

Name	Runnable_TCS_TimeManagement
Type	Fixed
Inter-runnable Variables	EA_LocalTimeClock EA_SubCounter
Managed Input Interfaces	N/A
Managed Output Interfaces	TCS_SWC_Clock
Per Instance Memory	N/A
Return	void
Description	This runnable is responsible for: • Keeping a local clock source to maintain the time between Master Clock Signal receptions • Providing a local clock source

[$SDD_TCS 009]
Runnable_TCS_PermitExecutionShutdown shall monitor PermitExecution input in order to execute.
[$Satisfies $SRS_TCS 009]

[$SDD_TCS 024]
Runnable_TCS_PermitExecutionShutdown shall set TCS_SUSD_ShutdownPermission to Enable when the ECU Reset bit in the SUSD_TCS_PermitExecution value is FALSE after checking that there are no pending NVM writes.
[$Satisfies $SRS_TCS 013]

[$SDD_TCS 010]
Runnable_TCS_PermitExecutionShutdown shall allow execution of all functions when the Wakeup(s) Active bit in the PermitExecution interface is TRUE and not set to FALSE even when wakeup(s) is no longer active.
The below image shows the state machine of Runnable_TCS_PermitExecutionShutdown.



[$Satisfies $SRS_TCS 010]

[$SDD_TCS 032]
Runnable_TCS_PermitExecutionShutdown shall write the RAM mirror block SumTCSInitValue into NVM during the power-down process. Write operation shall be prevented if the battery voltage is not good to perform it, by checking BatteryVoltage_DisableNVMWrite from SUSD.
[$Satisfies $SRS_TCS 023]

[$SDD_TCS 005]
SumTCSInitValue should be retrieved before starting the SUM_TCS module and use to substitute the last received value following initialization.
[$Satisfies $SRS_TCS 005]

[$SDD_TCS 007]
EA_LocalTimeClock shall be initialized with the value of the RAM mirror block SumTCSInitValue which is retrieved from NvM during initialization of the SUM_TCS module.
[$Satisfies $SRS_TCS 007]

Name	`Runnable_TCS_PermitExecutionShutdown`
Type	Fixed
Inter-runnable Variables	EA_LocalTimeClock EA_SubCounter
Managed Input Interfaces	SUSD_TCS_PermitExecution RP_SUSD_TCS_BatteryVoltage_DisableNVMWrite
Managed Output Interfaces	TCS_SUSD_ShutdownPermission TCS_SWC_Clock
Per Instance Memory	N/A
Return	`void`
Description	This runnable is responsible for: • Handling execution and shutdown permissions • Only once when first allowing execution: • Initializing `EA_LocalTimeClock` and `TCS_SWC_Clock` by reading `SumTCSInitValue` from NVM • Initializing `EA_SubCounter` to zero

[$SDD_TCS 033]
When the configuration parameter `TcsGenerateInitRunnable` is set to TRUE, the initialization is done by the `Runnable_TCS_Init` that calls the static `TCS_Init`.
[$Satisfies $SRS_TCS 020]

Name	`Runnable_TCS_Init`
Type	Fixed
Inter-runnable Variables	EA_LocalTimeClock
Managed Input Interfaces	N/A
Managed Output Interfaces	TCS_SWC_Clock
Per Instance Memory	N/A
Return	`void`
Description	This runnable is responsible for for calling the `TCS_Init` function.

4.5.2 Triggered Runnables

[$SDD_TCS 008]
Runnable_TCS_Dem_Clock is called each time a local clock is needed for DEM time stamping.
[$Satisfies $SRS_TCS 008]

Name	Runnable_TCS_Dem_Clock
Type	Event
Inter-runnable Variables	EA_LocalTimeClock
Managed Input Interfaces	N/A
Managed Output Interfaces	TCS_Dem_Clock
Per Instance Memory	N/A
Return	Std_ReturnType
Description	This runnable is responsible for: • Providing the timestamp to Dem when needed for a DTC event

[$SDD_TCS 021]
TCS_TotalVehicleTimeRead is called each time the Total Vehicle Time value is needed.
[$Satisfies $SRS_TCS 014] [$Satisfies $SRS_TCS 024] [$Satisfies $SRS_TCS 025]

Name	TCS_TotalVehicleTimeRead
Type	Event
Inter-runnable Variables	N/A
Managed Input Interfaces	N/A
Managed Output Interfaces	N/A
Per Instance Memory	N/A
Return	Std_ReturnType
Description	This runnable is responsible for: • Providing the Total Vehicle Time value

Name	TCS_TotalVehicleTimeCheckRead
Type	Event
Inter-runnable Variables	EA_LocalTimeClock
Managed Input Interfaces	N/A
Managed Output Interfaces	N/A
Per Instance Memory	N/A
Return	Std_ReturnType
Description	Function for condition check read of Total Vehicle Time value.

4.6 Runtime Scenario

4.6.1 Initialization

SUM_TCS has two states, `INIT` and `RUN`. The `INIT` state is entered during startup and when the initialization function has been called. The `RUN` state is entered only when the `PermitExecution` interface flags are set to enable. There is no shutdown state, rather each runnable shall monitor the `PermitExecution` flag to determine when to begin shutting down the module.

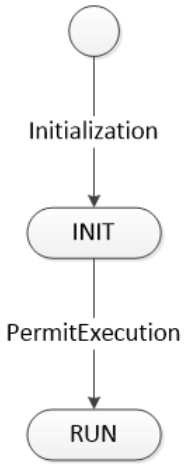


Figure 4.1: SUM_TCS state diagram

4.6.2 Runnables and communication

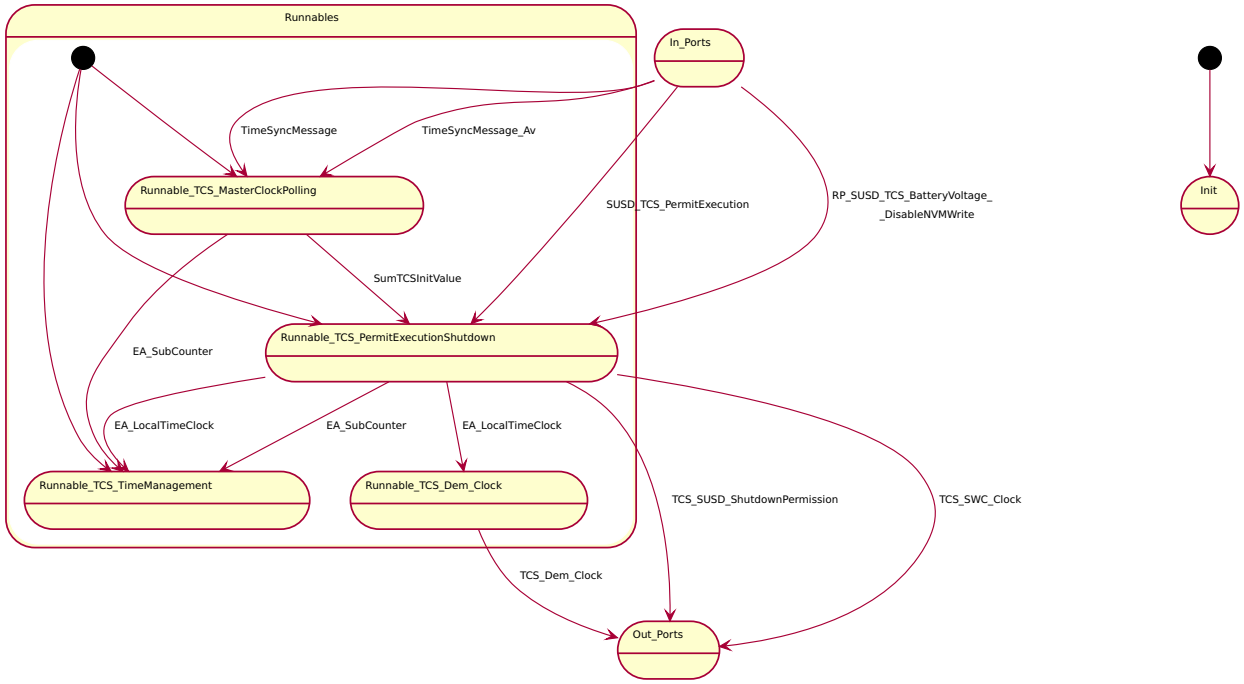


Diagram 4.1: Overview of TCS communication

5 Modification protocol

Version	Author	Date	Description of Changes
3.1.0	SUM Development Team	17/11/2020	First release of TCS as per GB8002
3.2.0_Beta3	SUM Development Team	12/18/2020	Added Runnable_TCS_PermitExecutionShutdown state machine, Updated section 4.5.1
3.2.1	SUM Development Team	18/03/2021	Release of RTA-SUM_TCS 3.2.1 (RTA-SUM 1.1.0)
3.2.2	SUM Development Team	09/06/2021	Release of RTA-SUM_TCS 3.2.2 (RTA-SUM 1.2.0). Updated section 3.1.2
3.2.4	SUM Development Team	17/09/2021	Release of RTA-SUM_TCS 3.2.4 (RTA-SUM 2.0.0)
3.2.5	SUM Development Team	01/02/2021	Updated section 3.1.2
3.3.0	SUM Development Team	12/04/2022	Release of RTA-SUM_TCS 3.3.0 (RTA-SUM 2.2.0)
8.0.0	SUM Development Team	22/08/2024	First version of RTA-SUM TCS for SDV 2.0.
8.0.1	SUM Development Team	10/12/2024	First version of RTA-SUM TCS for SDV 2.0. Updated Sections 2.1, 2.3, 4.5.1, 4.6.2 and 2.4.2 Updated Sections related to the DID 0xF129 : 3.1.2 and 4.5.2. Updated Sections related to initialization: 3.1.2, 4.1 and 4.5.1. Updated Section 4.3.2. Updated sections 3.1.2 and 4.5.1.